SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

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Predictive Analytics

CA3 Project Report

FOOD DELIVERY TIME PREDICTION AND ANALYSIS USING PREDICTIVE ANALYTICS

COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

Food delivery services depend on timely delivery to provide a good customer experience. Delivery time can vary due to factors such as traffic density, weather conditions, delivery distance, vehicle condition and order-related information. Predicting delivery time can help estimate when an order may reach a customer.

This project, titled *Food Delivery Time Prediction and Analysis Using Predictive Analytics*, uses historical food delivery data to estimate delivery duration in minutes. The project follows the Knowledge Discovery in Databases (KDD) process, including data understanding, data cleaning, feature engineering, exploratory data analysis, data preprocessing, model building and evaluation.

Three regression algorithms were implemented: Linear Regression, Decision Tree Regressor and Random Forest Regressor. The models were evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). A chronological data split was used to evaluate model performance on later delivery records.

In the current notebook results, Random Forest achieved the lowest test MAE and RMSE among the three models. The project demonstrates how predictive analytics can be applied to a practical problem using a structured workflow and interpretable evaluation metrics.

Keywords: Predictive Analytics, KDD, Food Delivery, Regression, Linear Regression, Decision Tree, Random Forest, Machine Learning

1. INTRODUCTION

1.1 Background

Food delivery platforms have become an important part of the food service industry. Customers expect their orders to arrive within a reasonable time, and delivery time is an important part of their overall experience. However, delivery duration is not always fixed. It may vary depending on traffic, weather, distance between the restaurant and customer, vehicle condition and other delivery-related factors.

An inaccurate delivery-time estimate may create inconvenience for customers and make delivery planning more difficult. Predictive analytics can help address this problem by using historical data to identify patterns and estimate the delivery duration of new orders.

Predictive analytics uses statistical techniques and machine learning algorithms to predict an unknown or future outcome from available data. In this project, historical food delivery records are used to build regression models that estimate delivery time in minutes.

1.2 Problem Statement

Food delivery time depends on several factors, including delivery distance, traffic density, weather conditions, vehicle condition and delivery-person information. The objective of this project is to predict food delivery time in minutes using historical delivery data.

The project treats the task as a regression problem because the target variable, Time_taken(min), is a continuous numerical value.

The prediction is intended to be made after a delivery person has been assigned but before the order is picked up.

1.3 Objectives

The main objectives of this project are:

1. To understand and analyse the supplied food delivery dataset.
2. To clean the dataset and handle missing or invalid values.
3. To create useful features, including delivery distance and order-time information.
4. To explore relationships between delivery time and selected variables through visualizations.
5. To train Linear Regression, Decision Tree Regressor and Random Forest Regressor models.
6. To compare the models using MAE, MSE, RMSE and MAPE.
7. To select a suitable model using validation performance.
8. To generate an estimated delivery time for a sample new order.

1.4 Scope of the Project

The project focuses on estimating food delivery duration using historical delivery records. It covers data preparation, exploratory analysis, feature engineering, model development, model comparison and prediction.

The project is implemented in Python using Jupyter Notebook and common data science libraries, including Pandas, NumPy, Matplotlib and Scikit-learn.

The project is an academic prototype. It does not connect to a live food delivery platform, track delivery persons in real time or guarantee the actual arrival time of an order.

1.5 Significance of the Project

This project demonstrates how historical data can be converted into useful predictions through a systematic analytical process. Comparing multiple regression models also helps identify which approach provides more accurate estimates on unseen delivery records.

The project provides practical experience in data preprocessing, regression modelling, model evaluation and interpretation, which are important areas of Predictive Analytics.

2. LITERATURE REVIEW

2.1 Yalçinkaya and Areta Hızıroğlu (2024)

Yalçinkaya and Areta Hızıroğlu [1] conducted a comparative study of machine learning models for predicting food delivery time. The researchers used delivery-related information such as order time, delivery location, weather conditions, traffic intensity and delivery-person details. They compared Linear Regression, Decision Tree, Random Forest and XGBoost models using evaluation measures such as MAE, RMSE and R^2. Their findings indicated that XGBoost performed better than the other evaluated models. The study highlights the importance of comparing different algorithms and examining feature importance when developing a delivery-time prediction system. It is closely related to the present project, which compares Linear Regression, Decision Tree Regressor and Random Forest Regressor using historical delivery records.

2.2 Garg et al. (2025)

Garg et al. [2] presented a study titled "Food Delivery Time Prediction in Indian Cities Using Machine Learning Models." The researchers examined delivery-time prediction in Indian urban environments using contextual and geographical information. Their work considered factors such as traffic density, weather conditions, local events, restaurant and customer coordinates, and delivery-person information. Several algorithms, including Linear Regression, Decision Trees, Random Forest, XGBoost and LightGBM, were compared. Their reported results showed that LightGBM achieved an R^2 score of 0.76 and an MSE of 20.59. The study emphasizes the importance of data preprocessing, feature selection and contextual information for improving prediction performance. The present project follows a similar approach but focuses on three comparatively simple regression algorithms that are easier to understand and explain.

2.3 Liu et al. (2021)

Liu et al. [3] studied on-time last-mile delivery and order assignment using travel-time predictors. Their research addressed the importance of estimating travel time when making delivery-assignment decisions. Accurate travel-time predictions can support better allocation of orders and improve the planning of last-mile delivery operations. The study demonstrates that delivery-time estimation is not only useful for informing customers but can also support operational decision-making. In the present project, historical delivery information is used to predict delivery duration. Although the project does not perform order assignment or route optimization, it applies the related principle of using predictive models to estimate delivery time.

2.4 Hildebrandt and Ulmer (2022)

Hildebrandt and Ulmer [4] investigated supervised learning methods for estimating arrival times in restaurant meal delivery. Their work focuses on the use of data-driven prediction to estimate when restaurant orders will arrive. The study is relevant because restaurant delivery times are affected by multiple operational factors and may vary across individual orders. Supervised learning provides a way to learn patterns from historical delivery records and use them to

estimate future arrival times. The present project applies this general approach by training three regression models on historical food delivery data and evaluating their performance on later records.

2.5 Zhu et al. (2020)

Zhu et al. [5] examined order-fulfilment cycle-time estimation for on-demand food delivery. Their research addresses the need to estimate the time required to fulfil an order in a delivery environment where operational conditions may vary. Order-fulfilment time estimation is important for improving service planning and providing useful time estimates. This work is relevant to the present project because both focus on estimating delivery-related time using data-driven methods. The current project uses a simpler experimental setup based on data cleaning, feature engineering, chronological data partitioning and comparison of three regression algorithms.

2.6 Summary of the Literature Review

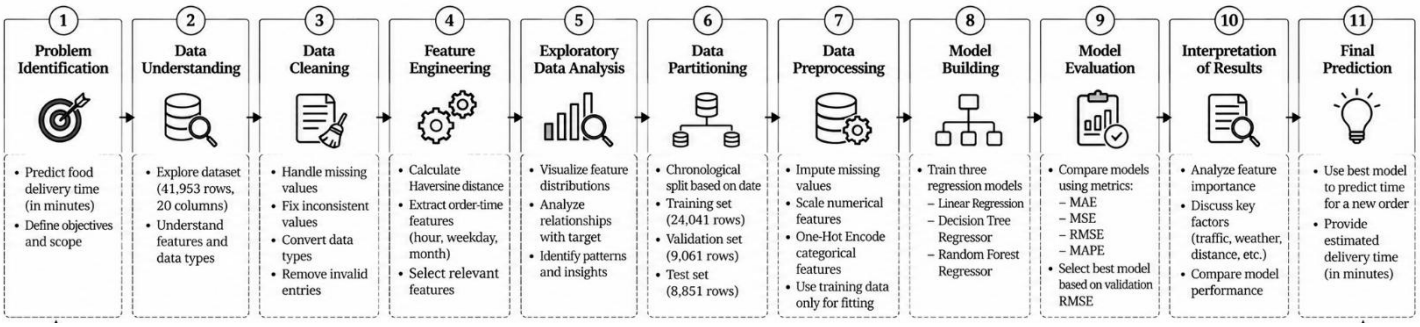
The reviewed studies show that machine learning is useful for delivery-time estimation and related logistics decisions. Previous research has examined traditional regression models, decision trees, ensemble learning, travel-time prediction and order-fulfilment time estimation. The studies also highlight the importance of contextual variables, geographical information, data preprocessing and appropriate evaluation methods.

Based on these findings, the present project implements an end-to-end KDD workflow using a historical food delivery dataset. Linear Regression is used as a baseline, while Decision Tree and Random Forest are included to evaluate whether tree-based models provide more accurate predictions. The models are compared using MAE, MSE, RMSE and MAPE, with chronological splitting used to evaluate performance on later delivery records.

3. WORKFLOW AND METHODOLOGY

3.1 Overview of the KDD Workflow

The project follows the major stages of the Knowledge Discovery in Databases process. Each stage contributes to converting the original delivery dataset into a trained and evaluated prediction model.



3.2 Problem Identification

The first stage was to define the problem clearly. The project aims to predict food delivery duration in minutes using historical order and delivery information.

The target variable is Time_taken(min). Since this variable represents a numerical duration, regression algorithms were selected.

The intended prediction point is after a delivery person has been assigned but before pickup. Therefore, the model should use only information that is appropriate for this prediction stage.

3.3 Data Understanding

The supplied dataset contains 41,953 records and 20 columns. The records cover the period from 11 February 2022 to 6 April 2022.

The dataset contains information related to delivery persons, restaurant and delivery locations, weather conditions, traffic density, vehicle details, order information and delivery duration.

The original dataset was supplied as a .txt file containing JSON data. Python's JSON library was used to load the records, and Pandas was used to convert them into a DataFrame for analysis.

During this stage, the dataset's dimensions, column names, data types, missing values, duplicate records and summary statistics were examined.

3.4 Data Cleaning

Raw data may contain inconsistent values and formatting issues. Data cleaning was performed before model development.

The main cleaning operations included:

- Removing unnecessary spaces from text values.
- Standardizing missing-value markers such as NaN, None and null.
- Correcting inconsistent category spelling, including the city value Metropolitian.
- Removing extra text from the target field where necessary.
- Converting numerical columns to numeric data types.
- Handling invalid delivery-person ratings and other unsuitable values.
- Converting order dates and order times into datetime formats.
- Removing duplicate records and records without a usable target or order date.

These operations helped make the data more consistent and suitable for analysis.

3.5 Feature Engineering

Feature engineering was performed to create useful information from the existing columns.

3.5.1 Delivery Distance

The dataset contains the latitude and longitude of the restaurant and delivery location. These coordinates were used to calculate the straight-line distance between the two locations using the Haversine formula.

The formula estimates the distance between two points on a sphere using their geographic coordinates. The result was stored in a new feature called distance_km.

This feature provides an approximate measure of geographical separation. It is not the same as actual road distance because roads, routes and traffic are not represented by the calculation.

3.5.2 Order-Time Features

The order time and order date were used to create the following features:

- Order hour: The hour at which the order was placed.
- Order weekday: The day of the week on which the order was placed.
- Order month: The month in which the order was placed.

These features allow the models to learn possible time-related patterns in delivery duration.

3.6 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of delivery times and examine their relationships with selected variables.

Five exploratory graphs were created, as shown in Figures 2–6.



Figure 2. Distribution of Food Delivery Time

This histogram shows how delivery durations are distributed across the dataset. It helps identify common delivery-time ranges and the overall spread of the target variable.

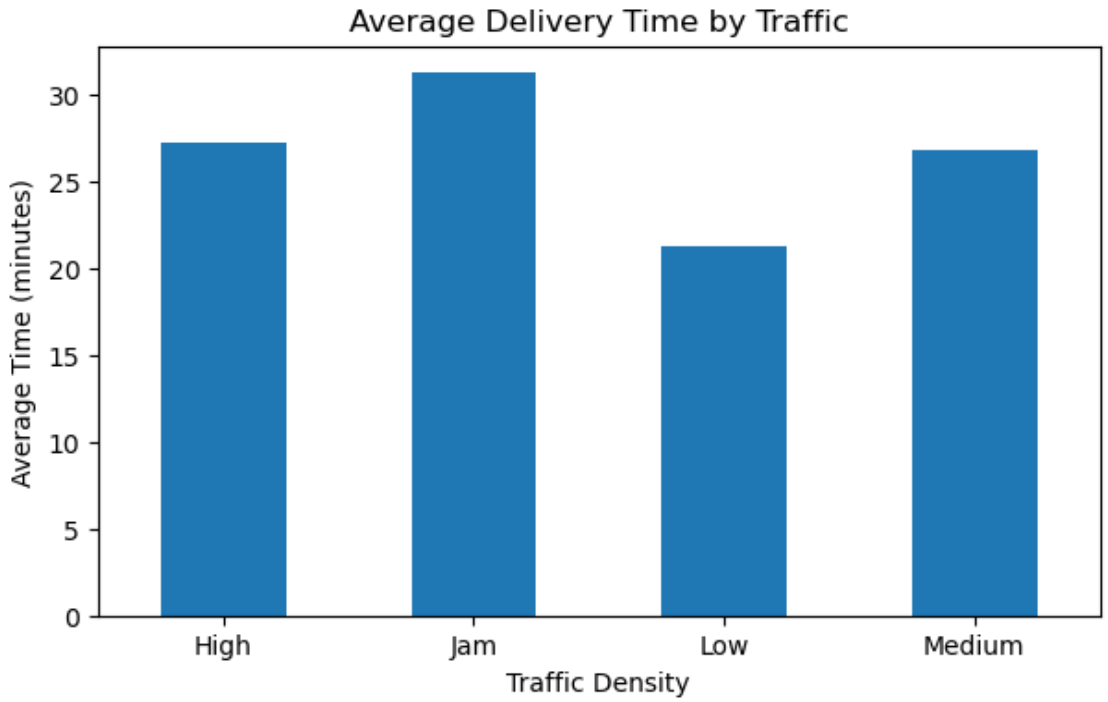


Figure 3. Average Delivery Time by Traffic Density

This bar chart compares average delivery time across traffic categories. It helps examine whether average delivery duration differs between traffic conditions.

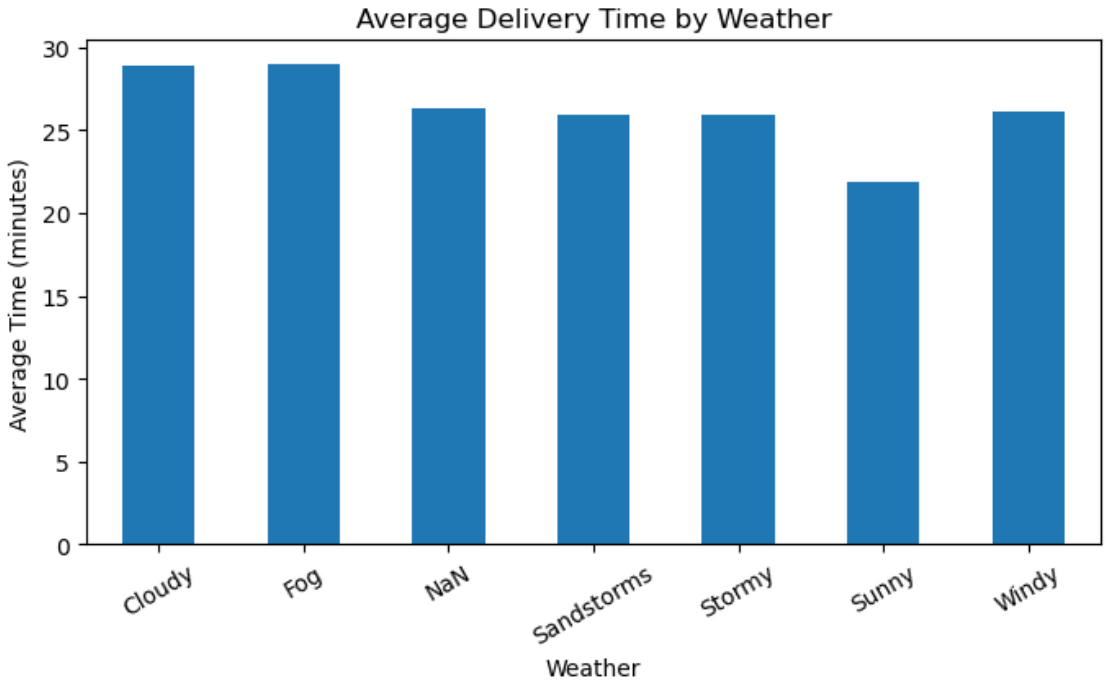


Figure 4. Average Delivery Time by Weather

Figure 4 compares average delivery time across the recorded weather categories. The “NaN” bar represents entries without a valid weather category in the plotted data; it should not be interpreted as a type of weather. This original chart is retained without altering the source values.

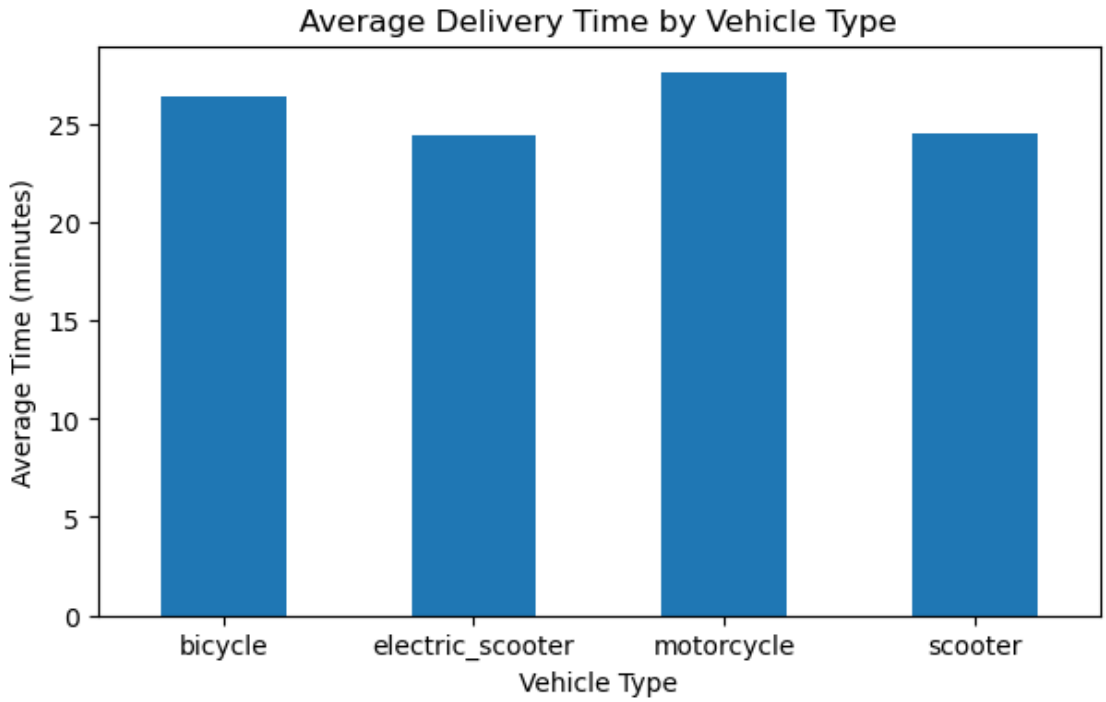


Figure 5. Average Delivery Time by Vehicle Type

This chart compares average delivery time for the vehicle types present in the dataset.

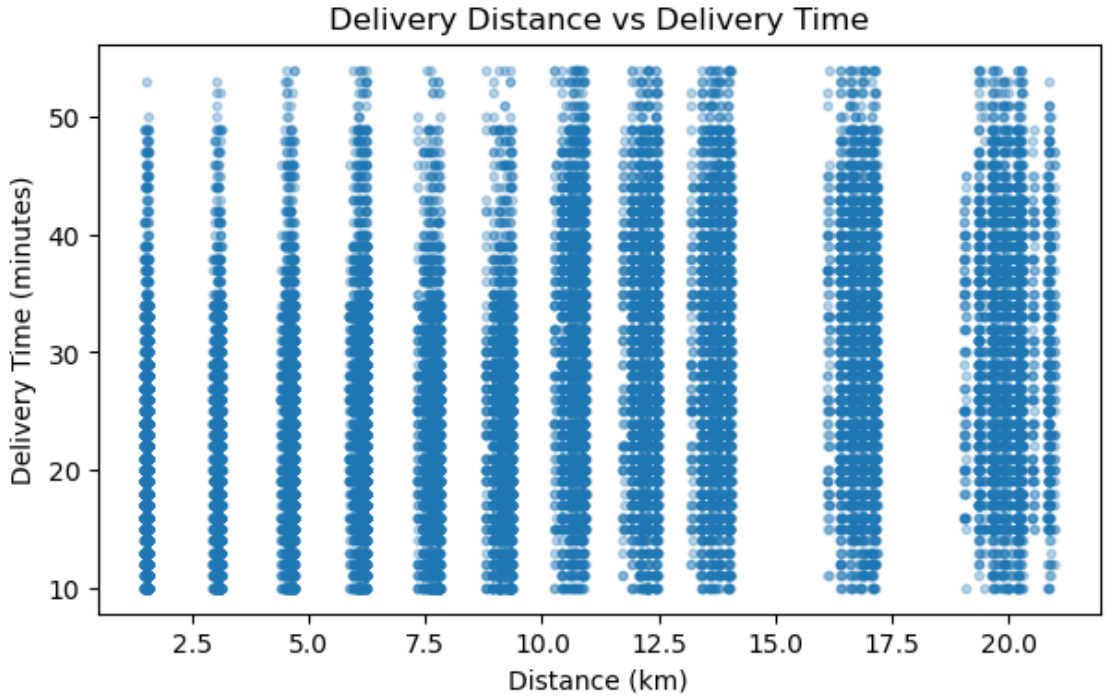


Figure 6. Delivery Distance vs Delivery Time

This scatter plot shows delivery duration against the calculated straight-line distance. It helps examine whether longer distances tend to be associated with longer delivery times.

The graphs are used for exploratory analysis. They show relationships in the available dataset but do not prove that any single factor causes a change in delivery time.

3.7 Data Partitioning

The dataset was divided chronologically into training, validation and test sets.

The chronological approach was selected because the project aims to predict delivery times for future orders. Training on earlier records and evaluating on later records provides a more realistic simulation than randomly mixing records across time.

The data was divided as follows:

Table 1. Chronological Dataset Partition

Dataset portion	Number of records	Purpose
Training set	24,041	Train the regression models
Validation set	9,061	Compare models and select the final model
Test set	8,851	Evaluate final performance on later unseen records
Total	41,953	Complete dataset

3.8 Data Preprocessing

Machine learning algorithms require the input data to be represented in a suitable numerical form. The following preprocessing operations were used.

3.8.1 Missing-Value Handling

Missing numerical values were replaced using the median of the training data. Missing categorical values were filled using the most frequent category in the training data.

Median imputation is less affected by extreme numerical values than mean imputation, while most-frequent imputation provides a simple method for handling missing categories.

3.8.2 Numerical Feature Scaling

StandardScaler was used to standardize numerical features. Scaling is particularly useful for Linear Regression when numerical inputs have different ranges, although tree-based models generally do not require scaling in the same way.

3.8.3 Categorical Encoding

Categorical features such as weather, traffic density, vehicle type and city were converted into numerical columns using One-Hot Encoding.

This process allows the regression models to use categorical information without assigning an artificial numerical order to category names.

3.8.4 Avoiding Data Leakage

The imputers, scaler and encoder were fitted using the training data only. The same fitted transformations were then applied to the validation and test sets.

This prevents information from the validation and test sets from influencing the preprocessing values learned during training.

Pickup time and record identifiers were excluded from the input features. The multiple_deliveries field was also excluded because its availability at the intended prediction time was unclear.

3.9 Model Building

Three regression algorithms were trained using the same training data and prepared features.

3.9.1 Linear Regression

Linear Regression was used as the baseline model. It estimates a numerical target using a linear relationship between input features and delivery time.

Its main advantage is simplicity. The results provide a baseline against which the performance of the other models can be compared.

3.9.2 Decision Tree Regressor

The Decision Tree Regressor learns decision rules from the training data. It divides records into groups based on feature values and predicts delivery time for each resulting group.

A maximum tree depth was used to limit the model's complexity. This helps reduce the risk of a single tree fitting the training data too closely.

3.9.3 Random Forest Regressor

The Random Forest Regressor combines predictions from multiple decision trees. The model was configured with 100 trees and a fixed random state for reproducibility.

By combining several trees, Random Forest can learn more complex patterns than a simple linear model while reducing dependence on the decisions of a single tree.

3.10 Model Evaluation and Selection

The three trained models were compared on the validation set using Root Mean Squared Error (RMSE), and the model with the lowest validation RMSE was selected for the example prediction. All three models were subsequently evaluated on the separate test set using MAE, MSE, RMSE and MAPE. The supplied report lists test metrics but does not provide the validation RMSE values, so the validation ranking cannot be independently checked here.

After model selection, all three models were evaluated on the test set using MAE, MSE, RMSE and MAPE. The test set was kept separate from the model-selection step.

This process makes it possible to compare model performance on later records that were not used for training.

3.11 Final Prediction

After model selection, a sample new delivery order was created with example values for delivery-person details, location coordinates, weather, traffic, vehicle type and order time.

The same preprocessing steps were applied to the new order, and the selected model generated an estimated delivery time in minutes.

The prediction demonstrates how the trained model can be used for a new input. Since the order values are illustrative, the output should be interpreted as an estimate rather than a guaranteed arrival time.

4. RESULTS AND DISCUSSION

4.1 Exploratory Data Analysis Results

The five EDA graphs provide a visual summary of the delivery dataset.

The delivery-time histogram shows the distribution and spread of the target variable. The traffic and weather charts allow average delivery duration to be compared across the available categories. The vehicle-type chart compares average delivery time by vehicle type, while the distance scatter plot examines the relationship between calculated geographical distance and delivery duration.

These patterns describe associations in the supplied dataset; they do not establish cause and effect. Figure 4 also includes a "NaN" weather category, indicating that some plotted weather values were missing or not assigned a weather label. The chart has been retained without changing the underlying dataset.

4.2 Model Evaluation Results

The three models were evaluated using the same test set. The saved notebook results are shown below.

Table 2. Test-Set Model Evaluation Results

Model	MAE	MSE	RMSE	MAPE (%)
Linear Regression	4.920	38.321	6.190	21.579
Decision Tree Regressor	3.521	19.997	4.472	15.128
Random Forest Regressor	3.356	17.567	4.191	14.410

4.3 Interpretation of Evaluation Metrics

Mean Absolute Error (MAE) is the average absolute prediction error in minutes. Mean Squared Error (MSE) averages squared errors in minutes squared. Root Mean Squared Error (RMSE) returns this squared-error measure to minutes and gives larger errors more weight. Mean Absolute Percentage Error (MAPE) expresses average absolute error as a percentage of the actual delivery time.

Linear Regression

Linear Regression produced a test MAE of approximately 4.920 minutes and an RMSE of approximately 6.190 minutes. It provides a simple baseline, but its errors were higher than those of the two tree-based models in this run.

Decision Tree Regressor

The Decision Tree Regressor achieved a test MAE of approximately 3.521 minutes and an RMSE of approximately 4.472 minutes. Its performance improved on the Linear Regression baseline, indicating that the tree-based model captured useful patterns in the data.

Random Forest Regressor

Random Forest achieved a test MAE of approximately 3.356 minutes and an RMSE of approximately 4.191 minutes. It also recorded the lowest MSE and MAPE among the three models in the saved results.

Its MAE indicates that the average absolute difference between predicted and actual delivery time was approximately 3.356 minutes on the test records. This does not mean every prediction was within that range; individual prediction errors may be smaller or larger.

4.4 Comparison of the Three Models

The comparison shows that both tree-based models performed better than Linear Regression on the test set in the current run. Random Forest achieved the lowest values for all four evaluation metrics.

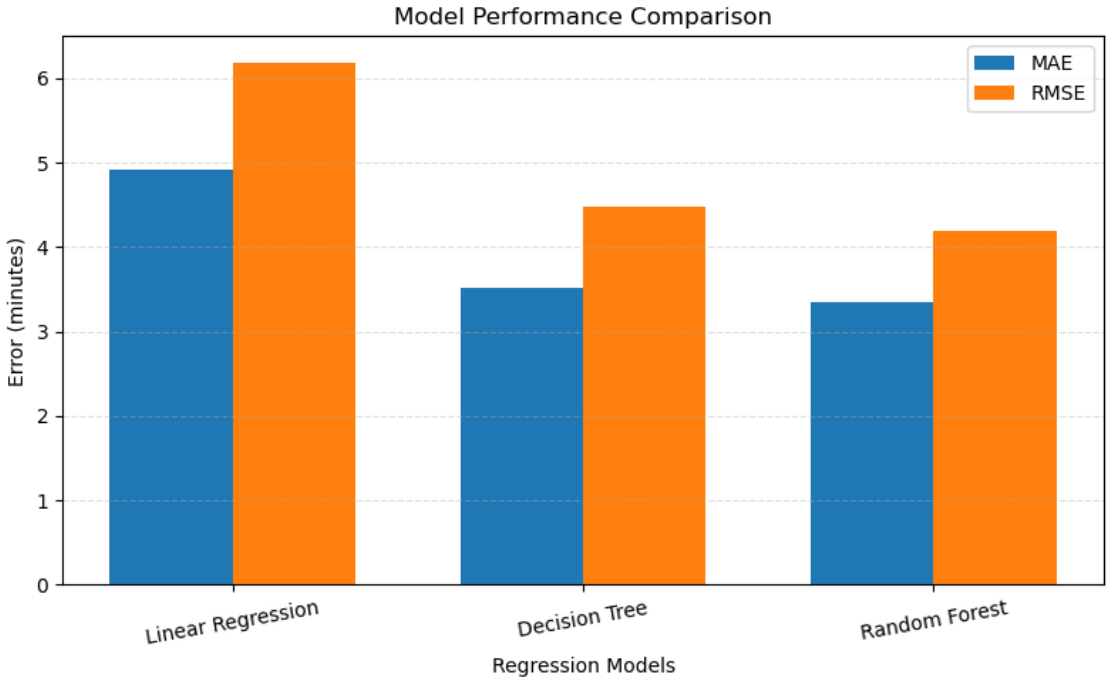


Figure 7. Model Performance Comparison Using MAE and RMSE

This result suggests that the relationships between the available delivery features and delivery time may not be represented fully by a simple linear model. The ensemble of trees used by Random Forest was more effective on this dataset in the current experiment.

The conclusion is limited to the supplied dataset, selected features, preprocessing steps and model settings. Other datasets or model configurations may produce different results.

4.5 Feature Importance

The notebook also calculates Random Forest feature importance and displays the five highest-ranked features. Their individual names and scores are not reproduced in the supplied report.

Feature importance indicates how much a feature contributes to the model's predictive decisions according to the model's importance calculation. It can help identify which input variables the model relies on most heavily.

However, feature importance does not prove that a feature causes delivery time to increase or decrease. It should be interpreted as a measure of contribution to the model, not as a causal relationship.

4.6 Key Findings

The main findings of the project are:

The KDD workflow provided a structured method for preparing the raw dataset and developing predictive models.

Feature engineering added straight-line delivery distance and time-based information to the original data.

Exploratory graphs helped examine delivery duration across traffic, weather and vehicle categories, as well as by distance.

The two tree-based models outperformed Linear Regression on the saved test results.

Random Forest achieved the lowest MAE, MSE, RMSE and MAPE in the current comparison.

The chronological split provided a more realistic evaluation of predicting later delivery records than a random split would.

The final prediction demonstrates how the trained model can estimate delivery duration for an example order.

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5. CONCLUSION

This project applied Predictive Analytics to the problem of food delivery time estimation. The work followed the KDD process, beginning with problem identification and data understanding, followed by data cleaning, feature engineering, exploratory data analysis, data partitioning, preprocessing, model building, evaluation and interpretation.

The supplied dataset contained 41,953 food delivery records and 20 columns. Data preparation included standardizing text values, handling invalid values, converting data types, calculating straight-line delivery distance and extracting order-time features. Numerical and categorical inputs were then prepared for machine learning.

Three regression algorithms—Linear Regression, Decision Tree Regressor and Random Forest Regressor—were trained and compared. MAE, MSE, RMSE and MAPE were used to evaluate

their predictions. In the saved results, Random Forest achieved the lowest test errors among the three models, making it the strongest performer in this experiment.

The project demonstrates that predictive analytics can be used to estimate delivery duration from historical data. It also shows the importance of data cleaning, feature engineering, fair model comparison and evaluation on unseen records.

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